

CY 512: Advanced Quantum Chemistry

Course Code	CY 512
Course Name	Advanced Quantum Chemistry
L-T-P-C	3-0-0-3
Prerequisites	B.Sc. (with Chemistry) or Teachers Consent
Intended for	UG/PG
Distribution	Core
Approval	8th Senate (Oct 2015)

Course Contents

- **Module I**
Review, Basic concepts and postulates of quantum mechanics of quantum mechanics: ScLecturesdinger wave equation, Eigenfunction, eigenvalues and operators, Interpretation of wavefunctions. [6 Lectures]
- **Module II**
Free particle, Particle-in-a-box, Rigid rotor, Harmonic oscillator, Hydrogen atom. [6 Lectures]
- **Module III**
Introduction to Self Consistent Field Theories; Valence Bond and Molecular Orbital theories; Introduction to Molecular spectroscopy: absorption, emission and resonance. [10 Lectures]
- **Module IV**
Approximate methods of quantum chemistry: variational principle; LCAO approximation; Huckel Theory; Time-independent perturbation theory. Many electron atoms: Orbital approximation, Slater determinant; Hartree-Fockselfconsistent field theory; Slater type orbitals. [6 Lectures]
- **Module V**
Angular momentum of many-particle systems. Spin orbital interaction; LS and JJ coupling. Spectroscopic term symbols for atoms. Molecules and Chemical bonding: Born-Oppenheimer approximation, MO and VB theories illustrated with H₂-molecule[8 Lectures]
- **Module VI**
Spectroscopic term symbols for diatomics; Directed valence & hybridization in simple polyatomic molecules. Elementary treatments of scattering and density functional theories [6 Lectures]

Text Books

1. Donald A. McQuarrie and John D. Simon, **Physical Chemistry: A Molecular Approach**, Viva Books, 1998.
2. Peter Atkins and Julio de Paula, **Physical Chemistry**, Oxford University Press, 7th Edition, 2002.

References

1. Ira N. Levine, **Quantum Chemistry**, Prentice-Hall, 1991.
2. Alberty and Silbey, **Physical Chemistry**, 4th Edition, Wiley, 2005.
3. Thomas Engel and Philip Reid, **Quantum Chemistry and Spectroscopy**, Pearson Education 2005.